

Abdelrahman Khalifa

Software Engineer — Rust · C/C++ · Linux · Open Source (GSoC 2026) · Codeforces Expert

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SUMMARY

Software engineer working in Rust, C and modern C++ on Linux and embedded systems. Two years of professional firmware and C++/Qt work at EKSON, a completed Google Summer of Code 2026 project with The Linux Foundation (OpenPrinting), and 7 PRs merged upstream in cups-rs and libcups, plus contributions to Zephyr RTOS, Boa and Rust Clippy. Codeforces Expert and ECPC 2023 finalist.

EXPERIENCE

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | May 2026 – Oct 2026

COSMIC desktop printer setup tool in Rust; mentors from OpenPrinting and System76 | [GSoC project page](#)

| github.com/Abd002/GSOC-2026 | github.com/Abd002/cosmic-printers

- Designed a printer-management stack as 5 Rust crates: `printers-core` (types, grouping queues into physical devices), `printers-client` (async varlink client), `printers-server` (CUPS, IPP, DNS-SD and Printer Application backend), `printers-ui` (iced/libcosmic) and `printers-app`.
- Exposed the backend over varlink IPC from `cosmic-settings-daemon`, so the C FFI code never loads in the Settings GUI process; the standalone app falls back to running the server in-process.
- Built a Printers page and an Add Printer flow for COSMIC Settings: printers grouped by device, state and supply levels, print queue control (pause, resume, cancel, move jobs), test pages and live updates; legacy printers are set up through Printer Applications (PAPPL) instead of PPD drivers.
- Ported cups-rs to libcups3 with CUPS 2 and CUPS 3 builds and CI; wrote FFI bindings for the libcups3 DNS-SD API; fixed crashes and a leaked client in libcups' Avahi DNS-SD code (C). **7 PRs merged upstream.**
- Tested on 5 real HP printers and Printer Applications (Ghostscript, Gutenprint, HPLIP, LPrint); CI checks formatting, Clippy and tests on several architectures and runs CUPS, Avahi and `ippeveprinter` to test discovery.

Embedded Systems Engineer — EKSON | Aug 2024 – Present

- Developed Zephyr RTOS firmware for an ESP32-S3 3-DoF motion simulator, migrating it from FreeRTOS for more deterministic behavior and production readiness.
- Integrated an MPU6050 IMU with calibration, filtering and compensation for stable motion estimation.
- Built a modern C++/Qt desktop control application with reliable UART control and optimized its performance by ~40% through asynchronous processing and fewer threads.

Embedded Linux Systems Intern — ODC | Dec 2023 – Jan 2024

- Built Linux systems with Yocto and Buildroot; worked with kernel modules, device drivers, device trees, file systems and hardware flashing.

Software Engineering Mentorship — Deloitte | 2024

- Selected for Deloitte's mentorship program on building mobile applications with React Native.

OPEN SOURCE

- **OpenPrinting cups-rs** (2026): IPP management operations, Printer Application support, CUPS 2 and CUPS 3 builds and CI, connections by IP/port, DNS-SD discovery, `lpoptions` support (PRs #18–#22, merged); earlier PRs #8, #9, #16.
- **OpenPrinting libcups** (2026): fixed crashes in `cupsDNSSDDelete()` when Avahi fails to initialize, and a leaked client in the Avahi DNS-SD code (#164, #165, merged).
- **COSMIC** (2026): Printers page and Add Printer flow in Settings ([cosmic-settings #2182](#)) and a varlink printers server in the settings daemon ([cosmic-settings-daemon #188](#)), in review.
- **Winter of Code 2026, OpenPrinting** (Jan 2026): implemented Rust versions of `lp` and `lpstat` on cups-rs (PR #11; also #9, #12, #13).
- **Boa JavaScript engine (Rust)** (Dec 2025): optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion; with tests (PR #4553).
- **Rust Clippy** (Dec 2025): fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).
- **Zephyr RTOS** (Mar 2024): implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules (150+ lines), merged upstream ([commit b117036](#)).

TECHNICAL SKILLS

Languages: Rust, C, C++ (modern C++, Qt), Python, Java, C#, SQL, JavaScript, Bash, Verilog/VHDL, MATLAB

Systems & Linux: Linux (Ubuntu/Debian), operating systems, FFI, IPC (varlink, D-Bus; zbus, zlink), async/event-driven design, multithreading, memory management, CUPS/IPP, DNS-SD/mDNS (Avahi), kernel modules and device drivers

Embedded: Zephyr, FreeRTOS, ARM Cortex-M (STM32, TM4C), ESP32, AVR, AUTOSAR, MISRA C, Yocto, Buildroot, QEMU; UART, SPI, I2C, CAN, LIN, USB, Ethernet

Tools & practices: Git/GitHub, Cargo, Clippy, CMake, Make, CI, unit and end-to-end testing (pytest, pytest-qt), debugging, code review, Xilinx, PCB design

Web, desktop & ML: React, Node.js, Express, MongoDB, REST APIs, ASP.NET Core, Flask; iced, libcosmic, Qt,

PROJECTS

- **AUTOSAR-Based V2X Ecosystem for SDV** (graduation project; FreeRTOS, AUTOSAR): software-defined vehicle prototype with ADAS (LKAS, driver monitoring), FOTA updates, and a secure bootloader with rollback. [GitHub](#)
- **IoT Device Communication System** (C++, Raspberry Pi 4, QEMU, Yocto): modular object-oriented framework for TCP unicast and UDP multicast between a Pi server and emulated clients; Yocto cross-compilation and Make. [GitHub](#)
- **Seat Heater Control System** (Tiva C, FreeRTOS): real-time controller with temperature monitoring, heater-level control and UART diagnostics, using tasks, mutexes and event flags. [GitHub](#)
- **AUTOSAR PORT and GPIO drivers** (C, TM4C): developed and validated MCAL drivers following the AUTOSAR specification.
- **Traffic Light Controller** (Verilog, Spartan-3E FPGA): FSM with debounce and delay modules, cutting signal timing errors by 20%. [GitHub](#)
- **Door Locker Security System** (C, two MCUs): password lock with keypad, LCD, I2C EEPROM, timer and DC motor; UART between the MCUs. [GitHub](#)
- **Operating System Simulator** (Java): CLI that simulates process scheduling, memory management, system calls and locks. [GitHub](#)
- **Function Solver and Plotter** (Python, PySide2, SymPy, Matplotlib): GUI that solves and plots two functions with intersection points; validation, error handling and end-to-end tests with pytest-qt. [GitHub](#)
- **Inventory Management System** (C#, WinForms, SQL, Telerik UI, Crystal Reports): stock, orders and deliveries with role-based access control, automated reports and local databases for offline use. [GitHub](#)
- **Sports Management Web App** (team; ASP.NET Core MVC, C#, MS SQL, Entity Framework): role-based CRUD dashboards; designed the relational schema and wrote the backend and account-management logic. [GitHub](#)
- **Phonebook** (React, Node.js, Express, MongoDB): full-stack app with a REST API. [GitHub](#)
- **Team projects (co-developed)**: Sarh Al-Tafawuq, a production web platform ([live](#)); Nafaqati, a Flutter Android expense tracker with on-device SMS import, charts and offline storage ([Google Play](#)).
- **More**: Children Safety web app (Flask, geolocation; [GitHub](#)), desktop chess in C# ([GitHub](#)), face-mask detection with a CNN, and a smart-garage IoT system deployed with Yocto on Intel Galileo.

ACHIEVEMENTS

- Codeforces Expert (handle `_Khalifa`) and ECPC 2023 finalist (Egyptian Collegiate Programming Contest).
- IEEE student chapter Treasurer: managed a budget of more than \$10k.

COURSES

Deep Learning (CNNs, TensorFlow/PyTorch) · Machine Learning Basics · RTOS Programming (FreeRTOS, OSEK/VDX, AUTOSAR OS) · Embedded Automotive (AUTOSAR, CAN/LIN, ISO 26262) · ARM Architecture (Cortex-M, drivers, DMA) · Embedded Systems Diploma (AVR, RTOS) · Full Stack Open (React, Node.js, MongoDB)

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | 2020 - 2025

Very Good (84.6%) with Honors.